

RAJ MAHESH SHARMA

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EDUCATION

Bachelor of Science (B.Sc.) in Computer Science, Yeshwantrao Chavan College
Relevant Coursework: Data Structures & Algorithms, SQL, OOP, Math, Statistics

Expected 05/2027
CGPA: 8.44/10

SKILLS

Programming & Databases	Python, SQL (MySQL)
AI & Core Competencies	Multi-Agent Orchestration, RAG, Predictive Modeling, Data Analytics
Frameworks & Libraries	FastAPI, LangGraph, OpenCV, PyTorch, Pandas, Numpy, Scikit-Learn
Tools & Infrastructure	Docker, ChromaDB, Git, GitHub, Jupyter Notebook

PROJECTS

Dynamic Multi-Agent Orchestrator | [GitHub](#)
Python, LangGraph, FastAPI, Docker, SQL, Server-Sent Events (SSE)

05/2026 - Present

- Architected a Dockerized multi-agent orchestrator using FastAPI and LangGraph, dynamically routing tasks across 4 specialized sub-agents (Decomposition, Retrieval, Critique, Synthesis).
- Integrated 4 distinct tool interfaces (including a Python sandbox and SQL database) with explicit failure contracts, streaming token-by-token responses in real-time via Server-Sent Events.
- Engineered a dynamic context manager for lossless data compression, alongside a custom 15-case adversarial evaluation pipeline featuring a self-improving meta-agent prompt loop.
- Deployed a 5-endpoint REST API with structured logging and asynchronous background workers, achieving zero-touch infrastructure setup via Docker Compose.

VidQuery: Multimodal AI Video Search Engine | [GitHub](#)
Python, OpenCV, YOLOv8, CLIP, ChromaDB, EasyOCR, PyTorch, SciPy

03/2026 - Present

- Engineered an end-to-end pipeline to search video footage using semantic text queries and zero-shot detection (YOLOv8).
- Architected a 1 FPS frame-extraction system via CLIP (ViT-B-32) and ChromaDB for sub-second vector retrieval.
- Developed cross-frame object tracking leveraging Intersection over Union (IoU) and the Hungarian algorithm.
- Integrated EasyOCR and K-Means clustering for dynamic text and dominant color indexing directly from frames.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate
Issued by Oracle · Validated expertise in Cloud and AI fundamentals, deploying ML workflows on OCI, and Responsible AI.

September 2025

LEADERSHIP EXPERIENCE

Chief Minister (Student Campus Lead)
Career Katta Initiative (MITSC)

07/2025 - Present

- Directed a 12-member team to spearhead the campus-level implementation of a state-wide skill development program.
- Orchestrated campus-wide campaigns and daily digital workshops for career guidance, engaging 100+ students.
- Acted as the primary liaison between college administration and the student body, facilitating access to industry experts.